

Weekly Progress Report

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Domain: Python Development Internship

Date of Submission: 07 July 2026

Week Ending: 03

I. Overview

This week, my main focus was to work on the goals I planned in Week 02. Instead of adding completely new projects, I focused on making my existing Python projects more professional and ready for final submission. I worked on project documentation, testing, deployment, and portfolio development. This week helped me understand how important testing and presentation are in real-world software development.

II. Achievements

1. Documentation & User Manual

- Created detailed user manuals for all four Python projects.
- Added installation steps, required libraries, execution process, and screenshots.
- Prepared proper documentation in PDF format so that users can understand project usage easily.

2. Python Project Contributions

Name of the Project:- URL Shortener (Week 03)

- Successfully deployed the URL Shortener project for public access using deployment tools.
- Added final UI improvements and validation for invalid URLs.
- Performed API testing using Postman to verify all endpoints.
- Improved database handling for better response speed.

Name of the Project:- File Organizer (Week 03)

- Added testing for large folders containing 500+ files.
- Verified custom category configuration.
- Improved backup handling before moving files.
- Reduced duplicate file processing time.

Name of the Project:- Password Manager (Week 03)

- Performed security testing on password storage.

- Verified bcrypt password hashing and encryption workflow.
- Tested export and import backup functionality.
- Improved error handling for invalid master password.

Name of the Project:- Quiz Game (Week 03)

- Added unit tests for scoring and timer logic.
- Improved difficulty-based score calculation.
- Expanded question bank with more programming questions.
- Fixed minor timer-related bugs.

3. Portfolio Website Development

- Started building a portfolio website to showcase all four projects.
- Added project descriptions, screenshots, skills, and GitHub links.
- Designed a clean UI using HTML, CSS, and JavaScript.

4. Unit Testing

- Learned basic testing using pytest.
- Wrote test cases for important project modules.
- Verified expected outputs and reduced runtime bugs.

III. Challenges & Hurdles

1. Deployment Issue on Render

Challenge:

While deploying the URL Shortener, I faced environment variable and configuration issues.

Resolution:

I checked deployment documentation carefully and fixed configuration step by step.

2. Writing Unit Tests

Challenge:

Writing pytest test cases was difficult because it was my first time working with testing frameworks.

Resolution:

I learned test structure from documentation and simple examples, then implemented tests gradually.

3. Portfolio Design Problems

Challenge:

Designing a responsive layout for the portfolio website took time.

Resolution:

I improved CSS styling and adjusted layouts for better presentation.

IV. Lessons Learned

This week taught me that software development is not only about writing code. Proper testing, documentation, deployment, and presentation are equally important.

I learned:

- Basic unit testing using pytest
- Project deployment workflow
- Importance of documentation
- Better frontend presentation skills

V. Learning Resources

- Pytest Documentation
- Render Deployment Docs
- Flask Documentation
- Stack Overflow
- ChatGPT

VI. Next Week's Goals

1. Complete portfolio website
2. Final optimization of all projects
3. Prepare final internship submission
4. Improve project UI and user experience

VII. Additional Comments

This week was different from previous weeks because I focused less on coding new features and more on testing, deployment, and documentation. I spent around 5–6 hours daily working on improvements. This helped me understand how real-world software projects are prepared before final delivery.